Aerodrome Race Optimisation

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CORO Project – 2025–2026

Problem Statement

Input

n aerodromes with coordinates (x_i, y_i), region $r_i \in \{0, \ldots, m\}$, start s, end t, minimum visits $V_{\min}$, range R. Distances:
$d(i,j) = \mathrm{round}\left(\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}\right)$.

Objective

Find path $s = v_1 \to \cdots \to v_k = t$ minimising $\sum_{\ell=1}^{k-1} d(v_\ell, v_{\ell+1})$ subject to:

- $d(v_\ell, v_{\ell+1}) \leq R$ (range constraint)
- Each region $\kappa = 1, \ldots, m$ has at least one visited aerodrome (coverage)
- $k \geq V_{\min}$ (minimum visits)

Difference from TSP

- Path from s to t, not a Hamiltonian cycle

Graph Construction

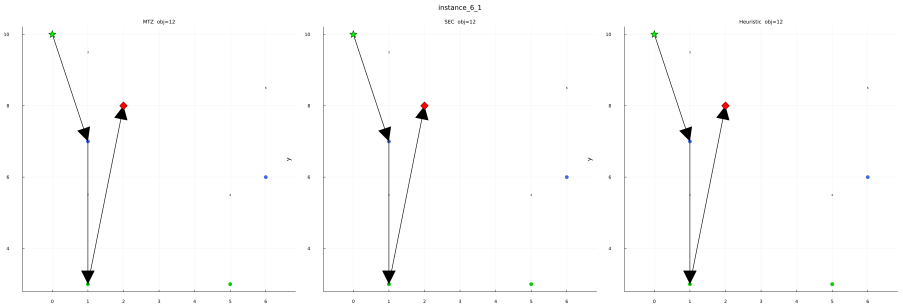
Arc set

Build directed graph $G = (V, A)$:

$$A = \{(i,j) : i \neq j,\ d(i,j) \leq R\}$$

Arcs violating range do not exist.

Reduces variables from $n(n-1)$ to $|A|$.



Example: `instance_6_1`

$n = 6,\ s = 1,\ t = 5,\ V_{\min} = 4,\ m = 2,$
$R = 6$

$d(1,3) = 3 \leq 6$ ✓ $\quad d(3,2) = 4 \leq 6$ ✓
$d(2,5) = 5 \leq 6$ ✓ $\quad d(1,2) = 7 > 6$

24 arcs within range.
Optimal: $1 \to 3 \to 2 \to 5$, cost $= 12$.

ILP Model 1 — MTZ Formulation

Variables

- $x_{ij} \in \{0,1\}$: arc (i,j) used
- $y_i \in \{0,1\}$: aerodrome i visited
- $u_i \in [0,n]$: position along path

Key constraints

C1–C4: flow conservation at s, t, intermediate nodes, force $y_s = y_t = 1$.

C5: $\sum_{i:\, r_i=\kappa} y_i \geq 1$ for each region κ

C6: $\sum_i y_i \geq V_{\min}$

C7 (MTZ): for $(i,j) \in A$, $i \neq s$, $j \neq s$:

$$u_i - u_j + n\, x_{ij} \leq n - 1$$

C8: $u_s = 0$; $u_i \leq n\, y_i$

How MTZ prevents subtours

If $x_{ij} = 1$: constraint forces $u_j \geq u_i + 1$.

Positions strictly increase along any used arc sequence.
A subtour would require positions to loop back — impossible.

Properties

- $O(n^2)$ constraints — polynomial
- Single solve, no iterations
- Weak LP relaxation: fractional u_i values allow subtour-like fractional solutions $\Rightarrow$ more B&B nodes

ILP Model 2 — Subtour Elimination Constraints

Same variables, no u_i

Replace C7–C8 with SEC.
For every $\emptyset \neq S \subseteq V \setminus \{s, t\}$:

$$\sum_{\substack{(i,j)\in A \\ i\in S,\, j\in S}} x_{ij} \leq |S| - 1$$

Among $|S|$ nodes, any acyclic subgraph has $\leq |S| - 1$ arcs. If $\geq |S|$ arcs exist inside S, a disconnected cycle exists.

Iterative solve-and-cut

1. Solve ILP with C1–C6 only.
2. BFS from s; find unreachable visited nodes U.

LP relaxation quality

$$z_{\mathrm{LP}}^{\mathrm{SEC}} \geq z_{\mathrm{LP}}^{\mathrm{MTZ}} \geq z_{\mathrm{ILP}}^{*}$$

SEC describes the subtour polytope exactly.

Tighter LP bound $\Rightarrow$ fewer B&B nodes pruned.

Tradeoff

- 2^{n-2} possible SEC constraints
- Separation: BFS in $O(n + |A|)$
- Our iterative approach re-solves from scratch at each iteration (overhead)

Heuristic — Three Phases

Phase 1: Greedy Construction

From s, at each step:

1. Find unvisited neighbours within R (not t).
2. If any lie in uncovered region $\rightarrow$ go to nearest such node.
3. Otherwise $\rightarrow$ go to nearest unvisited neighbour.
4. If no neighbour: Dijkstra escape on range graph.
5. Stop when all regions covered and $|\text{visited}| \geq V_{\min}$.
6. Connect to t directly or via Dijkstra.

Phase 2: 2-opt

Reverse interior segment $[v_i, \ldots, v_j]$ if:

Phase 3: Node Removal

For each interior node v_ℓ, remove if:

- $d(v_{\ell-1}, v_{\ell+1}) \leq R$
- $|\text{visited}| - 1 \geq V_{\min}$
- all regions still covered

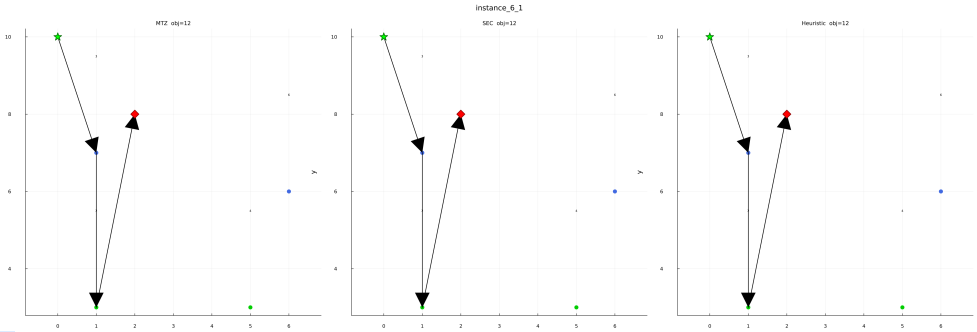
Remove node with largest distance saving. Repeat.

Known limitation

On sparse graphs (R tight), the Dijkstra escape skips already-visited intermediate hops to avoid duplicates, leaving direct connections that were never checked for range feasibility $\Rightarrow$ infeasible legs on large instances.

Results — Small Instance (instance_6_1)

Method	Obj.	Time (s)	B&B
MTZ	**12**	0.09	1
SEC	**12**	0.02	1
Heuristic	**12**	0.12	—



All three find optimal path $1 \to 3 \to 2 \to 5$, cost $= 12$.

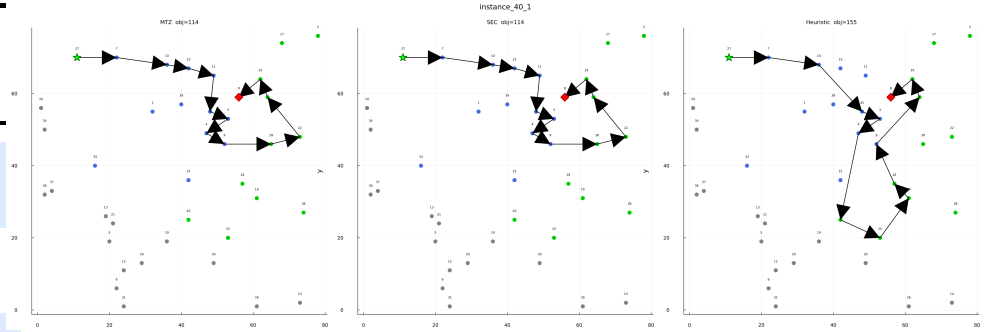
Both ILP models solve at root node (1 B&B node, no branching).

SEC added 3 cuts before converging.

Heuristic matches the optimum exactly.

Results — Medium Instance (instance_40_1)

Method	Obj.	Time (s)	B&B
MTZ	**114**	4.63	1 295
SEC	**114**	35.38	35
Heuristic	155	<0.01	—



Both ILP models prove optimal cost 114.

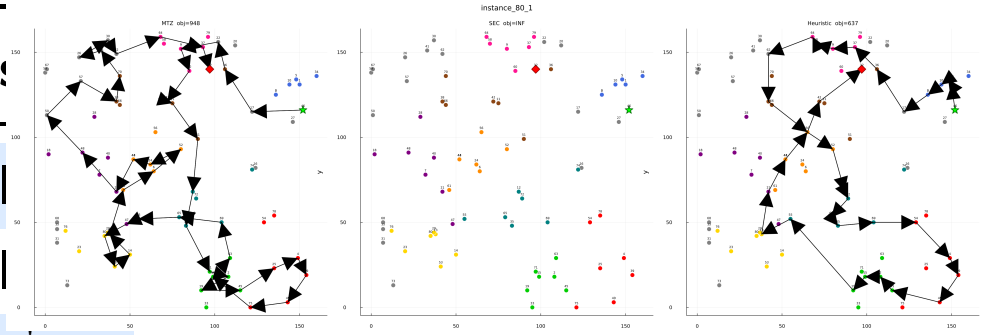
SEC uses 37× fewer B&B nodes (35 vs. 1 295) — tighter LP bound confirmed.

MTZ is 8× faster: iterative SEC discards the B&B tree after each cut batch (66 cuts total).

Heuristic gap: +35.96%.

Results — Large Instance (instance_80_1)

Method	Obj.	Time (s)	Status
MTZ	948	120.01	Time l
SEC	—	165.41	Time l
Heuristic	637	<0.01	Invalid †



† Heuristic leg 19 → 55: dist $= 56 > R = 31$.

MTZ found incumbent 948 (49 stops, 53 334 nodes) but could not prove optimality.

SEC added 105 cuts but found no valid solution within 165 s.

Each SEC re-solve consumed part of the time budget.

Results — Custom Instances Summary

Instance	n	MTZ obj.	MTZ nodes	SEC nodes	Heuristic	Gap
20_1	20	124 (opt.)	115 901	177	138	+11.3%
20_2	20	104 (opt.)	1 495	224	119	+14.4%
20_3	20	104 (opt.)	892	140	117	+12.5%
20_4	20	109 (opt.)	142 906	379	118	+8.3%
30_1	30	144 (opt.)	118 915	1 100	152	+5.6%
50_1	50	168 (opt.)	31 175	901	invalid	—
70_1	70	469 (t.l.)	32 330	11 647	invalid	—

- MTZ proved optimal on 5 of 7; SEC on 1 of 7 (20_3 only).
- SEC B&B nodes consistently 10×–376× fewer than MTZ.
- Heuristic feasible gaps: 5.6%–14.4% where valid.

Analysis — MTZ vs. SEC

Theory confirmed: SEC needs far fewer B&B nodes

$$z_{\mathrm{LP}}^{\mathrm{SEC}} \geq z_{\mathrm{LP}}^{\mathrm{MTZ}} \geq z_{\mathrm{ILP}}^{*}$$

Tighter LP bound $\Rightarrow$ more pruning by bound criterion (Cours 2).

But: iterative implementation is slower than MTZ

Re-solving from scratch after each cut batch discards the B&B tree.
Two sources of overhead: (i) root restart loses warm-start information; (ii) 90–134 cuts per instance, each triggering a full re-solve.

Key point: this is an implementation limitation, not a formulation weakness. Lazy-callback SEC would preserve the tree and outperform MTZ on all but tiny instances.

Connection to CORO lectures

- Cours 1 (LP duality): z_{LP} is a valid lower bound, exploited by B&B.
- Cours 2 (B&B): three pruning criteria — optimality, bound, infeasibility.

Analysis — Heuristic Failure on Sparse Instances

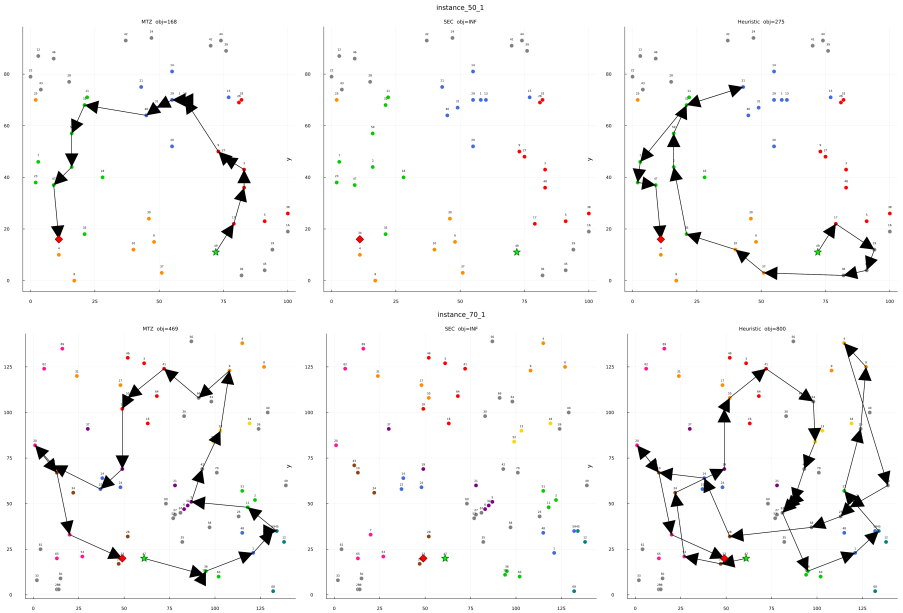
Root cause

Dijkstra escape returns a multi-hop sub-path, range-feasible by construction.

When the sub-path passes through an already-visited node, that node is skipped to avoid duplicates.

This leaves a direct connection between two non-adjacent nodes that was never checked for range feasibility.

Result: adjacent entries in the path vector exceed R.



Fix

Restrict Dijkstra to only unvisited intermediate nodes, even if this produces a longer sub-path.

Conclusion

Summary

- **MTZ**: practical winner in our implementation — proved optimal on 7 of 9 instances within 120 s. Simple, single solve, no iteration overhead.
- **SEC**: theoretically superior LP relaxation confirmed by 10×–376× fewer B&B nodes. Iterative re-solve overhead prevents it from outperforming MTZ in wall-clock time with our implementation.
- **Heuristic**: < 1 ms, gaps of 5–14% on feasible instances. Fails on sparse graphs due to range violation in Dijkstra escape.

What would improve results

- Implement SEC with **lazy callbacks** (single B&B run, no restarts).
- Use heuristic solution as **warm-start incumbent** for both ILP models.
- Apply **Lagrangian relaxation** on region-covering constraints (C5) for a tighter lower bound without SEC overhead (Cours 4).